Biology — Exam 12 — Chapters 1-7 (Full Syllabus), Detailed Marking Report

Seemab Ali | Class 9 | Annual Examination 2026 | Federal Board (FBISE) SSC-I | Paper Version 1 | Attempt date: 2026-07-22 | Marked: 2026-07-22 (corrected)

SECTION A (MCQS)	SECTION B (SHORT)	SECTION C (LONG)	TOTAL	GRADE
12/12	29/33	19.3/20	60.3/65 (92.8%)	A+

CORRECTED MARKING. An independent deep-check found the original marking (which reported 58.3/65, 89.7%, Grade A) had misread Section A. Seemab did not fill a bubble panel — she wrote her twelve MCQ answers as a plain numbered list on the last scan page, and the sloped ruled lines caused Q.9 and Q.12 to be misread as blank. Re-read at 400 DPI, all twelve are present and correct. Section A corrects from 10/12 to **12/12**; Section B (29/33) and Section C (19.3/20) are unchanged. A strong, well-rounded paper: a perfect MCQ score, only one under-developed OR answer, and two small "correct facts, missing the why" deductions.

SECTION A: Multiple Choice Questions — Score: 12/12

#	Question summary	Correct	Seemab	Result	Marks
1	First person to classify living organisms	C: Aristotle	C	Correct	1
2	Introduced 3-kingdom system, added Protista	B: Ernst Haeckel	B	Correct	1
3	Triplets of microtubules per centriole	C: Nine	C	Correct	1
4	Tissue culture agar medium hormone	D: Auxins	D	Correct	1
5	Organ system regulating blood calcium	C: Skeletal	C	Correct	1
6	First isolated nucleic acid material ("nuclein")	D: Friedrich Miescher	D	Correct	1
7	Diameter of the DNA double helix	B: 2 nm	B	Correct	1
8	Human enzyme optimum temperature range	C: 36-38°C	C	Correct	1
9	Trypsin's optimum pH	C: 8.00	C	Correct	1
10	Net ATP from complete oxidation of glucose	D: 36	D	Correct	1
11	Method of data analysis in biological method	B: ratio and proportion	B	Correct	1
12	Approx. number of specialized cell types from one zygote	C: 220	C	Correct	1

Section A: 12/12. Seemab recorded her MCQ answers as a plain numbered list ("MCQs:-") on the final page of her scan rather than a bubble panel. The ruled lines on that page slope, so each letter drifts upward relative to its number; read at 400 DPI, all twelve are legible: 1.C 2.B 3.C 4.D 5.C 6.D 7.B 8.C 9.C 10.D 11.B 12.C — every one matches the answer key, including Q.9 (trypsin's optimum pH, 8.00) and Q.12 (220 specialized cell types), both of which Seemab also wrote out correctly in her long-form answers elsewhere on this same paper.

SECTION B: Short Questions — Score: 29/33

Q.2(i), MAIN — Three criteria Carl Woese used to classify the three domains of life — 3/3

Main question

"Classification into 3 domains is based on the difference of the nucleotides in the rRNA (ribosomal Ribonucleic acid) of the cell, the cell's membrane lipid structure and it's sensitivity to antibiotics."

All three required criteria present: rRNA nucleotide sequence, membrane lipid structure, antibiotic sensitivity. Full marks.

Q.2(ii), MAIN — Label the plant cell diagram (A, B, C, D) — 3/3

Main question

A = Nucleus, B = Chloroplast, C = Vacuole, D = Cell wall.

All four labels correct, matching the key exactly (0.75 each). Full marks.

Q.2(iii), MAIN — Haematopoietic stem cell definition + any FOUR cell types — 3/3

Main question

"Haematopoietic stem cell is a type of stem cell that give rise to blood-related cell type. Any four types: Red blood cells, White blood cells, Platelets, Nerve cells."

Definition correct (1 mark). All four named cell types (RBC, WBC, platelets, nerve cells) are on the answer key's accepted list of six. Full marks.

Q.2(iv), MAIN — STEAM activity — onion skin slide preparation steps — 3/3

Main question

"(1) Place stem piece of onion skin in a drop of water on a slide and cover with a slip. (2) Observe under the microscope, first under the low power objective. (3) Then observe under high power objective and draw the diagram."

All three required steps present in correct order. Full marks.

Q.2(v), MAIN — Diploid/haploid chromosome numbers + why meiosis (not mitosis) forms gametes — 3/3

Main question

"Human somatic cells are diploid = 46 chromosomes, Gametes (sperm, egg) are haploid = 23 chromosomes. Meiosis halves the chromosomes number so that fertilization (23+23) restores 46 in the zygote, if gametes were formed by mitosis instead, they would carry 46 chromosomes each, and the number would double every generation."

Diploid/haploid numbers correct, and the full "why" reasoning (mitosis-formed gametes would double the chromosome count every generation) is spelled out completely. Full marks.

Q.2(vi), MAIN — Skeletal system organs + role in homeostasis — 3/3

Main question

"Organs: bones, cartilage, joints, tendons and ligaments. Role in homeostasis: Regulates the level of calcium and other minerals in the blood, by storing and releasing them from bones, also supports/protects the body and produces blood cells."

Organs and homeostatic role both complete and correct. Full marks.

Q.2(vii), MAIN — Who first isolated nucleic acid material, and how did the name "nucleic acid" come about? — 3/3

Main question

"Fedrick Miescher (swiss physician) isolated a whitish material from the nucleus of human pus cells and named it nuclein. Based on it's acidic properties, it is later renamed as nucleic acid."

Scientist, material name ("nuclein"), and the renaming reason (acidic properties) are all present. "Fedrick" is a phonetic misspelling of "Friedrich" but the person is unambiguous, not a content error. Full marks.

Q.2(viii), MAIN — RNA structure — strandedness, pentose sugar, replacement base, and why bases are "overhung" — 2/3

Main question

"RNA is single stranded. It's pentose sugar is ribose, the base that replaces thymine is uracil. The nitrogen bases are described as overhung."

Single-stranded, ribose, and uracil are all correctly stated (2 marks).

But the question specifically asks WHY the bases are called "overhung" — the answer only repeats the term itself rather than explaining that RNA (being single-stranded) has no complementary strand for its bases to bond to, unlike DNA. That reasoning was never written, so the third mark is not awarded. Lost 1 mark.

Q.2(ix), MAIN — Optimum pH for pepsin and trypsin, and why they differ — 2.5/3

Main question

"Pepsin works best in medium acidic in stomach at 2.00 pH and trypsin works best in medium alkaline in small intestine at 8.00 pH."

Both optimum pH values are correct (2.00 and 8.00), and their working locations (stomach vs small intestine) are correctly identified.

The location facts imply the reasoning, but the explicit "why" statement — that each enzyme's active site is shaped to function best at the specific pH of its own environment — is never spelled out. Half a mark held back for the missing explicit reasoning sentence.

Q.2(x), MAIN — ATP yield from glucose oxidation + daily ATP usage — 2.5/3

Main question

"Complete oxidation of 1 glucose molecule net 36 ATP molecules. An average human cell uses 100 to 150 of ATP per day."

36 ATP is correct, and the 100-150 range is correct.

The required unit "moles" was dropped ("100 to 150 of ATP" instead of "100 to 150 moles of ATP"). Half a mark lost for the missing unit on an otherwise numerically correct answer.

Q.2(xi) OR — How chewing food for longer helps digestion (amylase production; HCl production; pepsin activation) — 1/3

OR variant chosen

"Chewing food for longer produces more starch digesting enzymes amylase which mixes into the food and helps breaks down starch."

Only the first of three required points is present: chewing increases amylase production, which breaks down starch (1 mark).

The other two required points — that chewing also triggers HCl production in the stomach, and that this HCl activates the enzyme pepsin to break down protein — are entirely absent. The answer stops after one sentence. Two marks lost, the single largest deduction on the paper.

SECTION C: Long Questions — Score: 19.3/20

Q.3(i), MAIN — Labelled neuron diagram (cell body, dendrites, axon) + one function each — 5/5

Main question

Diagram correctly labelled: dendrites, cell body, axon. "Cell body: contains the nucleus, controls the neuron's activities. Dendrites: Conduct nerves impulse to the cell body. Axon: Conducts messages AWAY from the cell body."

Diagram fully correct with all three parts labelled (2 marks), and every function statement matches the key precisely, including the correct directionality for dendrites (impulses TO the cell body) and axon (messages AWAY from the cell body). Full marks.

Q.3(ii) OR — Comparison of any FOUR organ systems (organs + homeostatic role) — 5/5

OR variant chosen

Skeletal: bones, cartilage, tendons, joints, ligaments — regulates calcium/minerals in blood. Urinary: kidneys, ureters, bladder, urethra — regulates water, salts, H^+, wastes, pH. Lymphatic: lymph, lymph nodes, lymph vessels — maintains tissue fluid homeostasis. Muscular: skeletal, cardiac, smooth muscles — heat production for temperature regulation.

All four systems match the key's example table exactly, both organs and homeostatic role, for all four rows (1.25 marks each). Full marks.

Exam technique note: the question explicitly asks for the comparison "using a comparison table," and Seemab answered it correctly but as four headed paragraphs instead of an actual table. Content is complete so marks are unaffected here, but board examiners can dock presentation marks for not following an explicit format instruction — worth practising this answer style as a real table (rows = systems, columns = organs / homeostatic role).

Q.3(iii), MAIN — DNA base-percentage calculation (cytosine = 34%) 5/5

Main question

"Cytosine = 34% since C≡G always pair equally, guanine = 34%. Remaining bases (A+T) = 100-68 = 32%, since A=T always, adenine = thymine = 16% each, this follows Chargaff's/Watson-Crick base pairing rule, adenine always equals thymine (2 hydrogen bonds), and cytosine always equals guanine (3 hydrogen bonds)."

Guanine = 34%, adenine = thymine = 16% each, and the full base-pairing rule with hydrogen-bond counts are all present and correct. Full marks.

Q.3(iv), MAIN — Effect of temperature, pH, and substrate concentration on enzyme reaction rate (Fig 7.3-7.5) 4.3/5

Main question

"Temperature: Rate increases with temperature up to the optimum (36-38° for human enzymes) then decreases sharply due to denaturation. pH: Each enzymes work best at specific pH called optimum pH. for example (trypsin and pepsin) are both protein digesting enzyme. Pepsin works best at medium acidic stomach at 2.00pH and trypsin works best in medium alkaline in small intestine at 8.00pH. Substrate Concentration: Rate increases proportionally with substrate concentration then plateaus once all active site are occupied called saturation of all active site."

Temperature (1.7/1.7) and substrate concentration (1.6/1.6) both fully and precisely match the key, including "saturation of active sites."

The pH part correctly names the optimum pH concept and gives the right worked example (pepsin 2.00, trypsin 8.00) but never describes the actual shape of the graph the question asks for — that the rate decreases sharply on either side of the optimum due to denaturation at extreme pH. 0.7 of the 1.7 pH marks held back for the missing graph-shape description.

Deduction Log (reconciles to headline total)

Question	Marks available	Marks lost	Reason	Marks awarded
Section A, Q.1-12	12	0	All twelve MCQs correct (corrected from an original misread of Q9/Q12 as blank)	12
Q.2(i)-Q.2(vii)	21	0	All fully correct	21
Q.2(viii)	3	1	Strandedness/sugar/base correct; the "why overhung" reasoning never written	2
Q.2(ix)	3	0.5	Both pH values + locations correct; explicit "active site shaped to environment" reasoning not stated	2.5
Q.2(x)	3	0.5	36 ATP and 100-150 range correct; unit "moles" dropped	2.5
Q.2(xi) OR	3	2	Only amylase-production point given; HCl production and pepsin activation never written	1
Q.3(i), Q.3(ii) OR, Q.3(iii)	15	0	All fully correct	15
Q.3(iv)	5	0.7	Temperature and substrate concentration parts perfect; pH part gives correct values but omits the sharp-decrease-at-extremes description	4.3
Total	**65**	**4.7**	**Reconciles exactly to headline**	**60.3**

Summary — Strengths and Areas for Improvement

Strengths

- Perfect Section A — 12/12: every MCQ correct, including Q.9 and Q.12, which were initially hard to pair against their answer letters because Seemab used a numbered list instead of a bubble panel on a page with sloped ruled lines; both confirmed correct at 400 DPI
- Section C was the strongest part of the paper (19.3/20, 96.5%) — the neuron diagram, the 4-organ-system comparison table, and the DNA base-percentage calculation all scored a clean 5/5
- Both diagram-labelling questions fully correct: the plant cell (nucleus, chloroplast, vacuole, cell wall) and the neuron (dendrites, cell body, axon), including correct directionality of nerve impulse flow
- The DNA percentage calculation was worked through flawlessly with the full Chargaff's/Watson-Crick reasoning, including the correct hydrogen-bond counts for each base pair
- Haematopoietic stem cell types (4 of 4 correct) and the diploid/haploid/meiosis explanation both scored full marks with complete reasoning, not just the bare facts
- 7 of 11 Section B questions scored a clean 3/3

Areas to Revise

- **Q.2(xi)OR stopped after one point of three:** amylase production covered, but chewing's role in triggering HCl production and activating pepsin was never written — the largest single deduction (2 marks)
- **"Why" reasoning omitted twice (Q.2viii, Q.2ix):** correct facts stated, but the causal explanation the question specifically asked for was left out both times
- **Q.3(iv) pH graph description incomplete:** correct optimum pH values given, but the actual shape of the rate-vs-pH curve (sharp fall on either side due to denaturation) was never described
- **Missing unit (Q.2x):** "100 to 150 of ATP" should read "100 to 150 moles of ATP per day"
- **Exam technique (Q.3(ii)OR):** the question explicitly asked for a comparison "using a table," and Seemab answered with four headed paragraphs instead. Content was complete so marks weren't affected, but practise writing this answer style as an actual table for board-exam presentation marks.

Comparison vs Previous Biology Attempts

Date	Exam	MCQ	Sec B	Sec C	Total	%	Grade
3 Apr 2026	exam-01-ch1-6	11/12	28/33	13/20	52/65	80.0%	A
30 Apr 2026	exam-02-ch7 (1st)	12/12	31/33	17.5/20	60.5/65	93.1%	A
2 May 2026	exam-02-ch7 (re-sit)	12/12	33/33	20/20	65/65	100%	A+
8 Jun 2026	exam-03-ch4	12/12	31/33	19.5/20	62.5/65	96.2%	A+
23 Jun 2026	exam-07-ch1-7	12/12	27/33	18.5/20	57.5/65	88.5%	A
29 Jun 2026	exam-08-ch1-7	12/12	25/33	17.5/20	54.5/65	83.8%	A
6 Jul 2026	exam-09-ch1-7	12/12	32/33	17.25/20	61.25/65	94.2%	A+
16 Jul 2026	exam-11-ch1-7	10/12	26.9/33	19.5/20	56.4/65	86.8%	A
22 Jul 2026	**exam-12-ch1-7 (this exam)**	**12/12**	**29/33**	**19.3/20**	**60.3/65**	**92.8%**	**A+**

This is a fresh, previously-unseen full-syllabus Ch1-7 paper (exam-12, Version 1). At 92.8% it clears the 90% A+ boundary and sits well above exam-11 (86.8%). Section C (19.3/20, 96.5%) continues Biology's recent strength in long questions, close behind exam-11's 19.5/20. Section B (29/33, 87.9%) is Biology's 2nd-best short-question result of the four comprehensive Ch1-7 papers (only exam-09's 32/33 is higher), held back mainly by the incomplete Q.2(xi)OR. Section A returned to a clean 12/12, showing exam-11's 10/12 dip was an isolated slip rather than a developing pattern; Biology's MCQ record across all 9 attempts now stands at 105/108 (97.2%), with genuine errors only on exam-01 and exam-11.